

```
In [2]: import pandas as pd
import numpy as np
```

```
In [4]: data = pd.read_csv('CH.SC.U4CSE24156_EX-11.4_2(PCA).csv')

print(data.shape)
print(data.head())
```

```
(9, 4)
      Variable    PC1    PC2    PC3
0      Climate  0.190  0.017  0.207
1      Housing  0.544  0.020  0.204
2      Health  0.782 -0.605  0.144
3       Crime  0.365  0.294  0.585
4 Transportation  0.585  0.085  0.234
```

```
In [6]: print("Missing values:")
print(data.isnull().sum())
```

```
Missing values:
Variable    0
PC1         0
PC2         0
PC3         0
dtype: int64
```

```
In [8]: variables = data['Variable']

pc1 = data['PC1']
pc2 = data['PC2']
pc3 = data['PC3']

print("Variables:")
print(variables)

print("\nPC1:")
print(pc1)

print("\nPC2:")
print(pc2)

print("\nPC3:")
print(pc3)
```

Variables:

```
0      Climate
1      Housing
2      Health
3      Crime
4      Transportation
5      Education
6      Arts
7      Recreation
8      Economy
```

Name: Variable, dtype: object

PC1:

```
0    0.190
1    0.544
2    0.782
3    0.365
4    0.585
5    0.394
6    0.985
7    0.520
8    0.142
```

Name: PC1, dtype: float64

PC2:

```
0    0.017
1    0.020
2   -0.605
3    0.294
4    0.085
5   -0.273
6    0.126
7    0.402
8    0.150
```

Name: PC2, dtype: float64

PC3:

```
0    0.207
1    0.204
2    0.144
3    0.585
4    0.234
5    0.027
6   -0.111
7    0.519
8    0.239
```

Name: PC3, dtype: float64

```
In [12]: pc1_index = pc1.abs().idxmax()

print("Variable with highest loading in PC1:")
print(data.loc[pc1_index, 'Variable'])

print("Loading value:")
print(data.loc[pc1_index, 'PC1'])
```

Variable with highest loading in PC1:

Arts

Loading value:

0.985

```
In [14]: pc2_index = pc2.abs().idxmax()

print("Variable with highest loading in PC2:")
print(data.loc[pc2_index, 'Variable'])

print("Loading value:")
print(data.loc[pc2_index, 'PC2'])
```

Variable with highest loading in PC2:
Health
Loading value:
-0.605

```
In [16]: pc3_index = pc3.abs().idxmax()

print("Variable with highest loading in PC3:")
print(data.loc[pc3_index, 'Variable'])

print("Loading value:")
print(data.loc[pc3_index, 'PC3'])
```

Variable with highest loading in PC3:
Crime
Loading value:
0.585

```
In [18]: print("Principal Component Analysis Result")
print("-----")

print("PC1 :", data.loc[pc1_index, 'Variable'],
      "(", data.loc[pc1_index, 'PC1'], ")")

print("PC2 :", data.loc[pc2_index, 'Variable'],
      "(", data.loc[pc2_index, 'PC2'], ")")

print("PC3 :", data.loc[pc3_index, 'Variable'],
      "(", data.loc[pc3_index, 'PC3'], ")")
```

Principal Component Analysis Result

PC1 : Arts (0.985)
PC2 : Health (-0.605)
PC3 : Crime (0.585)

```
In [20]: result = pd.DataFrame({
    'Principal Component': ['PC1', 'PC2', 'PC3'],
    'Variable': [
        data.loc[pc1_index, 'Variable'],
        data.loc[pc2_index, 'Variable'],
        data.loc[pc3_index, 'Variable']
    ],
    'Loading': [
        data.loc[pc1_index, 'PC1'],
        data.loc[pc2_index, 'PC2'],
        data.loc[pc3_index, 'PC3']
    ]
})

print(result)
```

	Principal Component	Variable	Loading
0	PC1	Arts	0.985
1	PC2	Health	-0.605
2	PC3	Crime	0.585

In []: